

General selection criteria for each desktop manufacturing equipment is that the apparatus must be able to produce real parts from real production materials. For example, a desktop CNC mill must be able to machine materials such as brass and aluminium, as well as wax. Also, plastics production machinery can be used to fabricate parts large enough to be used for a number of applications. Those who are familiar with computerised numerical cutting know how great it can be for creating customised equipment and parts, hence, making it ideal for home offices.



Fig. 5: Desktop AOI machine

Welding time. Time taken for completion of welding of one board; expressed in minutes

Compatibility. Lead and lead-free reflow applications

Automatic or automated optical inspection, or AOI. This is an essential tool in an integrated electronics test strategy. It helps in fast and accurate inspection of PCBs to ensure that the PCBs leaving the production line have been assembled correctly and without manufacturing faults. The AOI machine is normally placed into the production line just after the soldering process, so that the defective card is removed from the assembly line at an early stage. The machine should be able to detect soldering and component defects. Specifications for evaluation and selection of an AOI are given below:

Throughput. Area scanning capability per second; number of components scanned per hour

Maximum board size. Maximum dimension that can be scanned; specified in length x breadth in millimetres

False calls. Percentage of false detection (<0.05% typical)

Component. Position, condition, wrong polarity, alignment skewed, leads bent or lifted and wrong value

Solder. Bridging, open, insufficient or short solder balls

Algorithms. Should be able to scan components based on colour and barcode

Programming skill level. Engineer, technician or operator

Operating system. Of integral computer

Assembled desktop 3D printer.

One area that remains largely unexplored is the use of additive manufacturing for electronics. Due to the complexity involved, development of 3D printers in the electronics industry is still in its infancy. Once convergence of electronics and 3D printing matures, it will have staggering implications on PCBs and rapid prototyping.

While it is unlikely that 3D printers for electronics will replace all traditional processes for in-house development of high-performance electronic device applications, these will be particularly useful in rapid prototyping.

Buying tips

Facilities offered by a machine need to be carefully analysed by matching these against costs to ensure that the choice made is correct for the particular production environment. General selection criteria and buying tips for selection, which are applicable to all equipment and processes, are as under:

Nature of business. Whether you want to be an original equipment manufacturer or contract/customer manufacturer, or both.

Profile of the product. Relates

to parameters of the board including size, complexity, components density and range of components to be mounted on the board; setup for manufacturing of a motherboard vastly differs from that of the control card of an AC.

Volume of production. Manually-operated machines are unsuitable for high-volume production. On the other side, fully-automated ones are expensive in terms of initial as well as running costs.

Quality of product. Machine and process should make the product meet industry or military standards.

Reliability. Once setup is placed into service, it is critical that it remains operational. Equipment in the setup therefore should withstand the rigors of production. Maintenance downtime should not be overlooked. Maintenance requirements should be compared carefully before selecting the equipment.

Life cycle. Machine should provide seven to ten years of service.

Vendor credibility. Credibility of manufacturer or vendor is of utmost importance. Their experience, standing in the market, provision of training facilities, clause of maintenance support, location of service centres, provision of field service and availability of spares are some aspects that must be considered.

Conclusion

Manufacturing setups for assembling PCBs are very expensive in terms of initial and running costs. These are available from highly-integrated as one setup to an ensemble of stand-alone machines. Once established, the facility must remain operative for all times except when under maintenance. It is therefore imperative that evaluation of machines is done on the basis of the nature of business, volume of production, profile of product and dependability of the vendor. **EEF**